# UML

## Cattle farm management system

**User**

- realName: string
- username: string
- password: string
- role: string
- backupWord1: string
- backupWord2: string

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User()
User(name: string, u: string, p: string, r: string)
+ getRealName(): string
+ getPassword(): string
+ getRole(): string
+ getBackupWord1(): string
+ getBackupWord2(): string
+ getUsername(): string
+ displayUserInfo(): void
+ saveToFile(): void
+ static loadUser(uname: string, userOut: User&): bool
+ generateBackupWord(): string

**LoginManager** *(inherits from User)*

LoginManager()
+ registerUser(): void
+ login(): void

**FinancialManager**

- records: FinancialRecord[MAX_FINREC]
- recordCount: int

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FinancialManager()
+ loadFinancialRecords(): void
+ recordExpense(): void
+ recordIncome(): void
+ calculateProfit(): void
+ generateFinancialReport(): void

**FinancialRecord**

- type: string
- category: string
- amount: float
- date: string

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FinancialRecord()
FinancialRecord(type: string, category: string, amount: float, date: string)
+ displayRecord(): void

**AnimalManager**

- cows: Cow[MAX_ANIMALS]
- bulls: Bull[MAX_ANIMALS]
- calves: Calf[MAX_ANIMALS]
- meatAnimals: MeatAnimal[MAX_ANIMALS]
- cowCount: int
- bullCount: int
- calfCount: int
- meatAnimalCount: int

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AnimalManager()
+ loadAllAnimalsFromFiles(): void
+ validateAnimalID(searchID: int): bool
+ addAnimal(): void
+ removeAnimal(): void
+ updateAnimal(): void
+ displayAllAnimals(): void
+ searchAnimalByType(): void
+ searchAnimalByID(id: int): void
+ findAnimalByID(searchID: int, type: string): int
+ locateAnimal(animalId: int, hour: int): void

**Animal**

- id: int
- name: string
- password: string
- gender: string
- ageYears: int
- category: string
- condition: string

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Animal()
Animal(id: int, name: string, gender: string, ageYears: int, category: string, condition: string)
+ displayInfo(): void
+ updateRecord(): void
+ updateRecord(): void
+ operator==(other: Animal&): bool
+ operator=(other: Animal&): Animal&
+ validateAnimalID( searchID: int ): bool
+ getID(): int
+ getName(): string
+ getGender(): string
+ getAge(): int
+ getCategory(): string
+ getCondition(): string
+ setID(newID: int): void
+ setName(newName: string): void
+ setGender(newGender: string): void
+ setAge(newAge: int): void
+ setCategory(newCategory: string): void
+ setCondition(newCondition: string): void

**TaskManager**

- tasks: Task[MAX_TASKS]
- taskCount: int
- am: AnimalManager

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TaskManager()
+ loadAllTasksFromFile(): void
+ validateTaskID(animalId: int): bool
+ addTask(): void
+ removeTask(): void
+ updateTaskStatus(): void
+ displayAllTasks(): void

**Task**

- animalId: int
- timeSlot: string
- taskType: string
- location: string
- description: string
- completed: string

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Task()
Task(animalId: int, timeSlot: string, taskType: string, location: string, description: string, completed: string)
+ displayTask(): void

parent / child

**Cow** *(inherits from Animal)*

- milkProduction: float
- breed: string
- breedingDate: string
- lastHeatDate: string
- nextHeatDate: string

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Cow()
Cow(id: int, name: string, gender: string, ageYears: int, condition: string, milkProduction: float, breed: string, breedingDate: string, lastHeatDate: string)
+ displayInfo(): void
+ updateRecord(): void
+ operator=(other: Cow&): Cow&

**Bull** *(inherits from Animal)*

- fertilityRate: float
- breedingSuccessRate: float

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Bull()
Bull(id: int, name: string, gender: string, ageYears: int, condition: string, fertilityRate: float, breedingSuccessRate: float)
+ displayInfo(): void
+ updateRecord(): void
+ operator=(other: Bull&): Bull&

**Calf** *(inherits from Animal)*

- breed: string
- DOB: string
- motherID: int
- fatherID: int

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Calf()
Calf(id: int, name: string, gender: string, ageYears: int, condition: string, breed: string, DOB: string, motherID: int, fatherID: int)
+ displayInfo(): void
+ updateRecord(): void
+ operator=(other: Calf&): Calf&

**MeatAnimal** *(inherits from Animal)*

- growthRate: float
- weight: float
- targetWeight: float
- requiredGrowthRate: float

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MeatAnimal()
MeatAnimal(id: int, name: string, gender: string, ageYears: int, condition: string, growthRate: float, weight: float, targetWeight: float)
+ displayInfo(): void
+ updateRecord(): void
+ operator=(other: MeatAnimal&): MeatAnimal&